Processing Power:
The Effects of Data Centers on Wolesale Electricity Markets

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Data centers are having and expected to have massive impact

- **This paper**: What has been (and will) be the impact of data centers on wholesale prices in the short run?
- **Method**: Dispatch model of the US.
- **Data**: Power plant characteristics/locations, demand data, data center locations/demand and forecasts (with assumptions).

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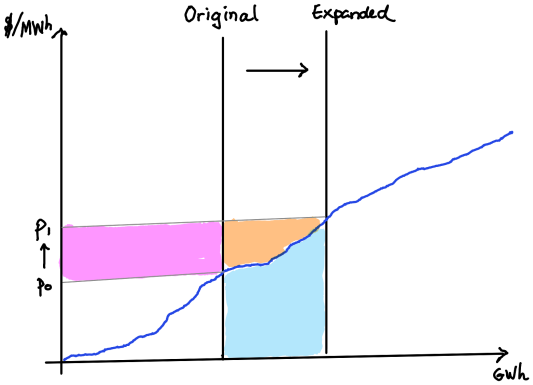
Two separate analyises

- **“Ex-post”**
  - Removing data center demand from baseline
- **“Forward looking”**
  - Adding more data center demand to baseline

Analogous approaches in methodology, but forward lookking considers different scenarios.

The effects of data centers are far from marginal



- For marginal demand increases, we can use the wholesale price directly to get generation cost.
- For *non-marginal* demand increases, we need a **model of the supply curve** to integrate out the effects.

The core piece: a "strawman" model

$$
\begin{aligned}
\min_{q_{it}} \quad & \sum_{i \in I_r} mc_{it} q_{it} \\
\text{s.t.} \quad & \sum_{i \in I_r} q_{it} = d_{rt} \quad \forall r, t \\
& 0 \leq q_{it} \leq K_{it} \quad \forall i \in I_r, t.
\end{aligned}
$$

- Minimize costs to serving inelastic demand with fossil fuel plants subject to other generators being held fixed.
- Shadow constraint on the market clearing constraint provides the marginal price of electricity.

Building on the shoulders of other "strawmen", but a bit bifferent

- Literature uses strawman model as an ideal (clearly unrealistic) model.
- Here model-to-model comparison may be cleaner for understanding impacts of changing conditions.
- But validation of the model is necessary to have more empirically valid predictions.

- Example:
  - As data centers push system, ignored bottlenecks become more prevalent.
  - Your approach is potentially a substantial underestimate.

Some low-hanging fruit that is relatively easy to incoporate

- I would bring in **optimization tools** if you have not already.
  - Adding realism to the model and testing its robustness is relatively straightforward.
  - Additional linear constraints that can incorporate ramping, batteries, wind/solar curtailment, and allow hydro to fluctuate (e.g., see Gonzales et al, Gowrisankaran et al.).
- I would validate the model with a calibration and/or estimation of some of the most relevant parameters.

Some potential for other calculations with additional price data

- The marginal price of electricity (and system lambdas) are observed for these markets (e.g., Borenstein and Bushnell).
- One could combine these prices with assumptions on the slope of the supply curve (based on your own data).
- One could look at more or less stringent frictions in the supply curve slope, e.g., based on observed frictions.

- If we use this **"marginal+ approach"**, what are the predictions? How do they compare?

Discussion part could be more enhanced, interesting!

- Engineers might have an edge at simulating impacts of growing data center demand.
- The discussion in the paper brings it back to economics: grid cost allocation, data center optimal location and its interactions with transmission.
  - Here need to be a bit careful with interpretation, e.g., with location of computes.
- Maybe this paper, maybe a future paper, but worth enhancing.

Thanks!

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